SEPARATION PROPERTIES OF A MORPHISM ARE CHECKED AFTER AN FPPF BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The identification of the diagonal of a base change with a base change of the diagonal. Invoked at the start of the counted proof (source lines 5996-6000). Reduces every separation property to a property of the diagonal.
- The criterion for checking a property of a representable morphism after a surjective flat finitely presented base change. Invoked inside the counted proof (source lines 6001-6015). Transfers the properties of the diagonal between the morphism and its base change.
- The definitions of the separation properties through the diagonal. Used inside the counted proof (source lines 6016-6030). Turns the diagonal properties back into the separation properties.

Source reference. Original lemma lemma-check-property-after-fppf-base-change: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5741–5741. Navigation pointer only: the counted text refers to the Stacks item lemma-check-property-after-fppf-base-change, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-properties-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 344–344. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-universally-injective: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2907–2907. Navigation pointer only: the counted text refers to the Stacks item lemma-universally-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-image-proper-is-proper: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6036–6036. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-image-proper-is-proper, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-image-universally-closed-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6022–6022. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-image-universally-closed-separated, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$, $g : \mathcal{Y} \to \mathcal{Z}$ be composable morphisms of algebraic stacks with composition $h = g \circ f : \mathcal{X} \to \mathcal{Z}$. If f is surjective, flat, locally of finite presentation, and universally injective and if h is separated, then g is separated.*

Proof. Consider the diagram

$$\begin{array}{ccccc} \mathcal{X} & \xrightarrow{\Delta} & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} & \longrightarrow & \mathcal{X} \times_{\mathcal{Z}} \mathcal{X} \\ & \searrow & \downarrow & & \downarrow \\ & & \mathcal{Y} & \longrightarrow & \mathcal{Y} \times_{\mathcal{Z}} \mathcal{Y} \end{array}$$

The square is cartesian. We have to show the bottom horizontal arrow is proper. We already know that it is representable by algebraic spaces and locally of finite type (Lemma lemma-properties-diagonal). Since the right vertical arrow is surjective, flat, and locally of finite presentation it suffices to show the top right horizontal arrow is proper (Lemma lemma-check-property-after-fppf-base-change). Since h is separated, the composition of the top horizontal arrows is proper.

Since f is universally injective Δ is surjective (Lemma lemma-universally-injective). Since the composition of Δ with the projection $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{X}$ is the identity, we see that Δ is universally closed. By Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-universally-closed-separated we conclude that $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{X} \times_{\mathcal{Z}} \mathcal{X}$ is separated as $\mathcal{X} \to \mathcal{X} \times_{\mathcal{Z}} \mathcal{X}$ is separated. Here we use that implications between properties of morphisms of algebraic spaces can be transferred to the same implications between properties of morphisms of algebraic stacks representable by

algebraic spaces; this is discussed in Properties of Stacks, Section stacks-properties-section-properties-morphisms. Finally, we use the same principle to conlude that $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{X} \times_{\mathcal{Z}} \mathcal{X}$ is proper from Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-proper-is-proper. □